

**Note:**

- During the attendance check a sticker containing a unique code will be put on this exam.
- This code contains a unique number that associates this exam with your registration number.
- This number is printed both next to the code and to the signature field in the attendance check list.

## Machine Learning

**Exam:** IN2064 / Retake

**Date:** Tuesday 31<sup>st</sup> March, 2026

**Examiner:** Prof. Dr. Stephan Günnemann

**Time:** 14:00 – 16:00

|   |     |     |     |     |     |     |     |     |     |
|---|-----|-----|-----|-----|-----|-----|-----|-----|-----|
|   | P 1 | P 2 | P 3 | P 4 | P 5 | P 6 | P 7 | P 8 | P 9 |
| I |     |     |     |     |     |     |     |     |     |

### Working instructions

- This exam consists of **16 pages** with a total of **9 problems**.  
Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 40 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources:
  - Two-sided DIN A4 sheet of handwritten notes (a print of digitally handwritten notes is allowed)
- **No other material (e.g. books, cell phones, calculators) is allowed!**
- Physically turn off all electronic devices, put them into your bag and close the bag.
- There is scratch paper at the end of the exam.
- Write your answers only in the provided solution boxes or the scratch paper.
- If you solve a task on the scratch paper, clearly reference it in the main solution box.
- All sheets (including scratch paper) have to be returned at the end.
- **Only use a black or a blue pen (no pencils, red or greens pens!)**
- **For problems that say “Justify your answer” you only get points if you provide a valid explanation.**
- **For problems that say “Derive” you only get points if you provide a valid mathematical derivation.**
- **For problems that say “Prove” you only get points if you provide a valid mathematical proof.**
- If a problem does not say “Justify your answer”, “Derive” or “Prove”, it is sufficient to only provide the correct answer.

Left room from \_\_\_\_\_ to \_\_\_\_\_ / Early submission at \_\_\_\_\_

## Problem 1 Weighted k-Nearest Neighbors (k-NN) (3 credits)

Recall that the  $\ell_p$ -distance between two points  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$  is defined as

$$d_p(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_p = \left( \sum_{i=1}^d |x_i - y_i|^p \right)^{1/p},$$

where the  $\ell_\infty$ -distance is based on the maximum norm and defined as  $d_\infty(\mathbf{x}, \mathbf{y}) = \max_{i=1, \dots, d} |x_i - y_i|$ .

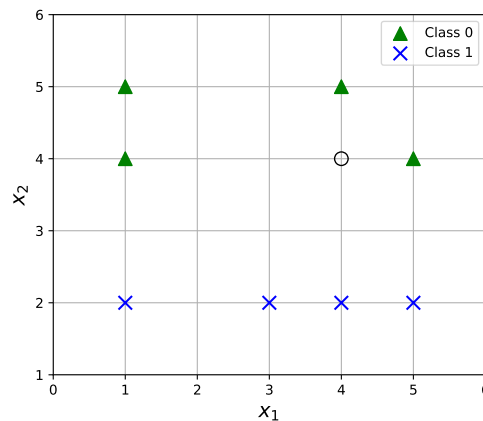
Moreover, the weighted k-NN classifier is defined as

$$\hat{y} = \arg \max_{c \in \mathcal{C}} \frac{1}{Z} \sum_{\mathbf{x}_i \in \mathcal{N}_k(\mathbf{x})} w_i \cdot \mathbb{I}(y_i = c) \quad \text{with} \quad \frac{1}{Z} = \sum_{\mathbf{x}_i \in \mathcal{N}_k(\mathbf{x})} w_i.$$

A potential issue in k-NN classification arises when multiple training points are equally distant to the point to be classified, making them all eligible as the k-th nearest neighbor. This can lead to ambiguity in the classification decision, as it is unclear which points should be considered in the majority vote.

One way to mitigate this is to locally enlarge the neighborhood to include *all equally distant* k-th nearest neighbors. For example in 1-NN: If there is a tie between two points (both at the same distance), then you would enlarge the neighborhood to include both points, effectively using two neighbors for the 1-NN decision.

We want to apply this k-NN variant with locally enlarged neighborhood to this binary classification problem:



- 0 ☐ a) Classify the test point (marked by  $\circ$ ) at coordinates (4, 4) using 3-NN with weights  $w_i = 1$  for the  $d_1$  and  $d_\infty$  distances. Justify your answer by showing the weighted votes for each class.

1 ☐

$p = 1$ : Class 0: 2 votes, Class 1: 1 vote  $\Rightarrow \hat{y} = 0$   
 $p = \infty$ : Class 0: 2 votes, Class 1: 3 votes  $\Rightarrow \hat{y} = 1$

b) Classify the test point (marked by  $\circ$ ) at coordinates (4, 4) using 3-NN with weights  $w_i = \frac{1}{d_p(\mathbf{x}, \mathbf{x}_i)}$  for the  $d_1$  and  $d_\infty$  distances. Justify your answer by showing the weighted votes for each class.

☐ 0  
☐ 1

$p = 1$ : Class 0: 2 votes, Class 1: 1/2 vote  $\Rightarrow \hat{y} = 0$   
 $p = \infty$ : Class 0: 2 votes, Class 1: 3/2 votes  $\Rightarrow \hat{y} = 0$

c) Consider any k-NN classifier with weights  $w_i = \frac{1}{d_p(\mathbf{x}, \mathbf{x}_i)}$  for  $p = 1$  and  $k \leq |\mathcal{D}|$  for the dataset  $\mathcal{D}$ . You are an adversary who may insert one additional training point  $\hat{\mathbf{x}}$  with the new label 2, with the goal of causing the test point ( $\circ$ ) to be misclassified as this new third class. Describe where you would place the new training point to achieve this. Justify your answer.

☐ 0  
☐ 1

Arbitrary close to (or directly at) coordinates (4,4) such that the weight of the new point  $\frac{1}{d_p(\mathbf{x}, \hat{\mathbf{x}})}$  exceeds the votes for the other class. Works because weighted k-NN classifier is sensitive to small distances.

## Problem 2 Probabilistic Inference (4 credits)

Suppose we observe i.i.d. data  $\mathcal{D} = \{x_1, \dots, x_N\}$ , with  $x_i \in \mathbb{R}$ , sampled from a shifted exponential distribution with unknown location parameter  $\theta \in \mathbb{R}$ :

$$p(x|\theta) = \begin{cases} \lambda \exp(-\lambda(x - \theta)) & \text{if } x \geq \theta \\ 0 & \text{otherwise,} \end{cases}$$

where  $\lambda > 0$  is known.

a) For which values of  $\theta$  is the data likelihood  $p(\mathcal{D}|\theta) = \prod_{i=1}^N p(x_i|\theta)$  zero? And for which values is it non-zero? Show your work.

Define indicator function:

$$\mathbf{1}(A) = \begin{cases} 1 & \text{if } A \text{ is true} \\ 0 & \text{otherwise.} \end{cases}$$

For each sample  $x_i$  we can write the likelihood as follows:

$$p(x_i|\theta) = \lambda \exp(-\lambda(x_i - \theta)) \mathbf{1}(x_i \geq \theta)$$

Thus,

$$p(\mathcal{D}|\theta) = \prod_{i=1}^N [\lambda \exp(-\lambda(x_i - \theta)) \mathbf{1}(x_i \geq \theta)]$$

If any  $x_i$  satisfies  $x_i < \theta$ , then the data likelihood is 0. Therefore, with  $m = \min\{x_1, \dots, x_N\}$ :

$$\begin{aligned} p(\mathcal{D}|\theta) &= \left[ \prod_{i=1}^N \lambda \exp(-\lambda(x_i - \theta)) \right] \mathbf{1}(\theta \leq m) \\ &= \lambda^N \exp(-\lambda \sum_{i=1}^N (x_i - \theta)) \mathbf{1}(\theta \leq m) \\ &= \lambda^N \exp(-\lambda \sum_{i=1}^N x_i + N\lambda\theta) \mathbf{1}(\theta \leq m) \end{aligned}$$

Therefore,

$$p(\mathcal{D}|\theta) \begin{cases} = 0 & \text{if } \theta > m \\ \neq 0 & \text{if } \theta \leq m. \end{cases}$$

b) Justify that the data likelihood is monotonically increasing in  $\theta$  in the region, where it is non-zero.

0  
1

$$p(\mathcal{D}|\theta) = \lambda^N \exp\left(-\lambda \sum_{i=1}^N x_i + N\lambda\theta\right) \mathbf{1}(\theta \leq m)$$

$N\lambda\theta$  is increasing in  $\theta$  because  $N\lambda > 0$ .

Therefore  $-\lambda \sum_{i=1}^N x_i + N\lambda\theta$  is also increasing in  $\theta$  because the added term is constant in  $\theta$ .

Because  $\exp(\cdot)$  is an increasing function,  $\exp(-\lambda \sum_{i=1}^N x_i + N\lambda\theta)$  is increasing.

Finally, because  $\lambda^N > 0$ ,  $p(\mathcal{D}|\theta)$  is increasing in the range where it is non-zero.

c) What is then the maximum likelihood estimate  $\theta_{\text{MLE}} = \arg \max_{\theta} p(\mathcal{D}|\theta)$ ? Show your work.

0  
1

To obtain the MLE, we need to maximize  $\ell(\theta) = \lambda^N \exp(-\lambda \sum_{i=1}^N x_i + N\lambda\theta)$  subject to  $\theta \leq m$ . Since  $\ell(\theta)$  is positive and monotonically increasing in  $\theta$  in the region where it is non-zero, its maximum is at the right boundary of this region:

$$\theta_{\text{MLE}} = m = \min\{x_1, \dots, x_N\}.$$

### Problem 3 Linear Regression (3 credits)

We want to perform ridge regression on a dataset consisting of  $N$  samples  $\mathbf{x}_i \in \mathbb{R}^D$  ( $D > 1$ ) with corresponding targets  $y_i \in \mathbb{R}$ , represented compactly as  $\mathbf{X} \in \mathbb{R}^{N \times D}$  and  $\mathbf{y} \in \mathbb{R}^N$ .

The closed-form solution for ridge regression is given by  $\mathbf{w}^* = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y}$ , where  $\lambda > 0$  is the regularization strength.

a) Show that  $\mathbf{w}^*$  can be equivalently written as  $\mathbf{w}^* = \mathbf{X}^T (\mathbf{X} \mathbf{X}^T + \lambda \mathbf{I})^{-1} \mathbf{y}$ , which is called the dual form of ridge regression.

*Hint: Use the fact that  $(\mathbf{X} \mathbf{X}^T + \lambda \mathbf{I})(\mathbf{X} \mathbf{X}^T + \lambda \mathbf{I})^{-1} = \mathbf{I}$ .*

We need to show that  $(\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T = \mathbf{X}^T (\mathbf{X} \mathbf{X}^T + \lambda \mathbf{I})^{-1}$ .

$$\begin{aligned} (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T &= (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \underbrace{(\mathbf{X} \mathbf{X}^T + \lambda \mathbf{I})(\mathbf{X} \mathbf{X}^T + \lambda \mathbf{I})^{-1}}_{\mathbf{I}} \\ &= (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} (\mathbf{X}^T \mathbf{X} \mathbf{X}^T + \lambda \mathbf{X}^T) (\mathbf{X} \mathbf{X}^T + \lambda \mathbf{I})^{-1} \\ &= \underbrace{(\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})}_{\mathbf{I}} \mathbf{X}^T (\mathbf{X} \mathbf{X}^T + \lambda \mathbf{I})^{-1} \\ &= \mathbf{X}^T (\mathbf{X} \mathbf{X}^T + \lambda \mathbf{I})^{-1} \end{aligned}$$

Therefore,  $\mathbf{w}^* = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y} = \mathbf{X}^T (\mathbf{X} \mathbf{X}^T + \lambda \mathbf{I})^{-1} \mathbf{y}$

b) When is the dual form useful from a computational perspective?

The primal form requires inverting a  $D \times D$  matrix, while the dual form requires inverting an  $N \times N$  matrix. When  $D \gg N$ , the dual form is therefore computationally more efficient.

## Problem 4 Linear Classification (4 credits)

Consider a binary generative classifier defined by class probabilities

$$p(y = 0) = \pi_0, \quad p(y = 1) = 1 - \pi_0,$$

and class-conditional densities given by Laplacian distributions

$$p(x|y = c) = \frac{1}{2b} \exp\left(-\frac{|x - \mu_c|}{b}\right),$$

where  $x \in \mathbb{R}$  is the input feature,  $\pi_0 \in [0, 1]$ ,  $b > 0$ ,  $c \in \{0, 1\}$ , and  $\mu_c \in \mathbb{R}$  are the class-conditional means.

a) Is the classifier linear in  $x$ ? Justify your answer.

We look at the decision boundary:

$$\begin{aligned} p(y = 1|x) &= p(y = 0|x) \\ \Leftrightarrow \frac{p(y = 1|x)}{p(y = 0|x)} &= 1 \\ \Leftrightarrow \frac{p(x|y = 1)p(y = 1)}{p(x|y = 0)p(y = 0)} &= 1 \\ \Leftrightarrow \frac{\frac{1}{2b} \exp\left(-\frac{|x - \mu_1|}{b}\right)(1 - \pi_0)}{\frac{1}{2b} \exp\left(-\frac{|x - \mu_0|}{b}\right)\pi_0} &= 1 \\ \Leftrightarrow \log \frac{\exp\left(-\frac{|x - \mu_1|}{b}\right)(1 - \pi_0)}{\exp\left(-\frac{|x - \mu_0|}{b}\right)\pi_0} &= 0 \\ \Leftrightarrow -\frac{|x - \mu_1|}{b} + \frac{|x - \mu_0|}{b} + \log \frac{1 - \pi_0}{\pi_0} &= 0 \\ \Leftrightarrow \frac{1}{b}(|x - \mu_0| - |x - \mu_1|) + \log \frac{1 - \pi_0}{\pi_0} &= 0 \end{aligned}$$

The decision boundary is not linear but piecewise linear.  
Therefore, the classifier is not linear.



b) Name one advantage of using a Laplace distribution instead of a Gaussian distribution for  $p(x|y)$ .

Some possible answers (other answers might be accepted):

- **Robustness to outliers:** Laplace is more robust because it penalizes deviations linearly, while Gaussian penalizes deviations quadratically.
- **Heavy-tailed:** Laplace is a better fit for data with extreme values.



## Problem 5 Convexity and the Hessian (6 credits)

*Multiple choice scoring rule:* Each sub-task is scored all-or-nothing. Full points are awarded if and only if exactly the correct set of options is selected. Any incorrect or missing selection yields 0 points for that sub-task.

a) Let  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  be twice continuously differentiable. Which of the following are correct?

- ☒  $f$  is convex if and only if  $f(\lambda \mathbf{x} + (1 - \lambda)\mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda)f(\mathbf{y})$  for all  $\mathbf{x}, \mathbf{y}$  and  $\lambda \in [0, 1]$ .
- ☒ If  $f$  is convex, then every local minimum is a global minimum.
- ☐ If  $\nabla^2 f(\mathbf{x})$  has all eigenvalues  $> 0$  at a single point  $\mathbf{x}_0$ , then  $f$  is strictly convex.
- ☒ The sum of two convex functions is always convex.
- ☐ If  $f$  is convex and  $g$  is convex, then  $f \circ g$  is convex.

b) Consider the function  $f(x, y) = x^2 - 2xy + y^2 + x$ . Which of the following are correct? (2 Points)

- ☐  $f$  is strictly convex.
- ☒  $f$  is convex but not strictly convex.
- ☐ The Hessian is positive semidefinite because all diagonal entries are positive.
- ☒ The Hessian is positive semidefinite because all eigenvalues are  $\geq 0$ .
- ☐  $f$  has a unique global minimum.

c) Let  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  be twice continuously differentiable and convex. A point  $\mathbf{x}^*$  satisfies  $\nabla f(\mathbf{x}^*) = \mathbf{0}$ . Which of the following are correct?

- ☒  $\mathbf{x}^*$  is a global minimum of  $f$ .
- ☐  $\mathbf{x}^*$  is the unique global minimum of  $f$ .
- ☐  $\nabla^2 f(\mathbf{x}^*)$  must be positive definite.
- ☒ There may exist other points  $\mathbf{x}' \neq \mathbf{x}^*$  with  $f(\mathbf{x}') = f(\mathbf{x}^*)$ .
- ☐  $\mathbf{x}^*$  is a saddle point.

d) Consider the function  $h(x, y) = e^{x+y}$ . Which of the following are correct? (2 Points)

- ☐ The Hessian of  $h$  has rank 2 everywhere.
- ☒ The Hessian of  $h$  has rank 1 everywhere.
- ☐  $h$  is strictly convex.
- ☒  $h$  is convex but not strictly convex.
- ☐  $h$  has a global minimum.

## Problem 6 Deep Learning (6 credits)

You are implementing a deep learning model to be trained with backpropagation. Let

$$z = Wx, \quad y = z + x, \quad s = y \odot y, \quad L = \sum_{i=1}^N s_i$$

where  $x \in \mathbb{R}^N$ ,  $W \in \mathbb{R}^{N \times N}$ ,  $z, y, s \in \mathbb{R}^N$  and  $L \in \mathbb{R}$ .  $\odot$  denotes elementwise multiplication. You have access to the following operations:

- `matmul(W, x)`: matrix-vector multiplication
- `add(u, v)`: elementwise addition
- `square(y)`: elementwise squaring
- `sum(s)`: summation over all entries

a) Draw the computational graph of the forward pass that computes  $L$  from  $W$  and  $x$ , using only the operations above.

A valid computational graph consists of:

- input nodes:  $W, x$
- intermediate nodes:  $z, y, s$
- output node:  $L$ ,

where

$$z = \text{matmul}(W, x), \quad y = \text{add}(z, x), \quad s = \text{square}(y), \quad L = \text{sum}(s).$$

with edges

$$W, x \rightarrow z \rightarrow y \rightarrow s \rightarrow L, \quad \text{and} \quad x \rightarrow y.$$

0  
1  
2

b) Write the chain rule expressions for  $\frac{\partial L}{\partial W}$  and  $\frac{\partial L}{\partial x}$  based on your computational graph using all intermediate variables.

$$\frac{\partial L}{\partial W} = \frac{\partial L}{\partial s} \cdot \frac{\partial s}{\partial y} \cdot \frac{\partial y}{\partial z} \cdot \frac{\partial z}{\partial W}.$$

$$\frac{\partial L}{\partial x} = \frac{\partial L}{\partial s} \cdot \frac{\partial s}{\partial y} \cdot \frac{\partial y}{\partial z} \cdot \frac{\partial z}{\partial x} + \frac{\partial L}{\partial s} \cdot \frac{\partial s}{\partial y} \cdot \frac{\partial y}{\partial x}$$

0  
1  
2

c) Which intermediate values in your computational graph must be cached during the forward pass in order to compute  $\frac{\partial L}{\partial W}$  and  $\frac{\partial L}{\partial x}$  during backpropagation without recomputation? **Justify your answer.**  
Note:  $W$  and  $x$  are assumed to be available and do not need to be cached.

- $\frac{\partial L}{\partial s_i} = 1$ : no stored values are required here
- $\frac{\partial s}{\partial y} = 2y$ :  $y$  must be cached.
- Computing  $\frac{\partial y}{\partial x}$  and  $\frac{\partial y}{\partial z}$ , where  $y = x + z$ , does not require additional stored intermediates.
- Computing  $\frac{\partial z}{\partial x}$  and  $\frac{\partial z}{\partial W}$  only requires  $x$  and  $W$ , which are assumed to be available.

The only required cached intermediate value is  $y$ .

0  
1  
2

## Problem 7 Support Vector Machines (3 credits)

*Multiple choice scoring rule:* Each sub-task is scored all-or-nothing. Full points are awarded if and only if exactly the correct set of options is selected. Any incorrect or missing selection yields 0 points for that sub-task.

Consider the soft-margin SVM primal problem:

$$\min_{\mathbf{w}, b, \xi} \quad \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^N \xi_i \quad \text{s.t.} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0, \quad \forall i,$$

where  $C > 0$  is a hyperparameter that controls the penalty for margin violations: each training point  $\mathbf{x}_i$  that does not satisfy the margin constraint  $y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1$  incurs a slack  $\xi_i > 0$ , and the total slack  $\sum_i \xi_i$  is penalized in the objective with weight  $C$ . A larger  $C$  penalizes violations more heavily.

The corresponding dual problem has Lagrange multipliers  $\alpha_i$  with  $0 \leq \alpha_i \leq C$ . At the optimum, the KKT conditions imply:

- $\alpha_i^* = 0$ :  $\mathbf{x}_i$  is correctly classified and lies outside or on the margin.
- $0 < \alpha_i^* < C$ :  $\mathbf{x}_i$  lies exactly on the margin boundary ( $\xi_i^* = 0$ ).
- $\alpha_i^* = C$ :  $\mathbf{x}_i$  may violate the margin ( $\xi_i^* \geq 0$ ).

For the special case  $C \rightarrow \infty$ , no margin violations are permitted and the problem reduces to the hard-margin SVM (which requires the data to be linearly separable).

a) A hard-margin SVM is trained on a linearly separable dataset  $\{(\mathbf{x}_i, y_i)\}_{i=1}^N$  in  $\mathbb{R}^2$ , yielding optimal dual variables  $\alpha_1^*, \dots, \alpha_N^*$ . Which of the following modifications to the training set will *not* change the optimal decision boundary?

- ☒ Removing a training point with  $\alpha_i^* = 0$ .
- ☒ Adding a new positive-class point that is correctly classified and lies outside the margin.
- ☐ Moving a support vector closer to the decision boundary along its perpendicular.
- ☐ Adding a new negative-class point that lies exactly on the positive margin boundary, i.e. at  $\mathbf{w}^{*\top} \mathbf{x} + b^* = +1$ .
- ☐ Adding a new positive-class point that lies exactly on the decision boundary, i.e. at  $\mathbf{w}^{*\top} \mathbf{x} + b^* = 0$ .

b) A soft-margin SVM with  $C = 10$  is trained. A particular training point  $\mathbf{x}_j$  with label  $y_j = +1$  has optimal dual variable  $\alpha_j^* = 10$  and slack  $\xi_j^* = 0.4$ . Which statements about  $\mathbf{x}_j$  are correct?

- ☐  $\mathbf{x}_j$  lies exactly on the margin boundary.
- ☒  $\mathbf{x}_j$  is correctly classified but lies strictly inside the margin.
- ☐  $\mathbf{x}_j$  is misclassified (on the wrong side of the decision boundary).
- ☐  $\mathbf{x}_j$  is not a support vector.
- ☐  $\mathbf{x}_j$  lies exactly on the decision boundary.

c) Two soft-margin SVMs are trained on the same non-separable dataset: one with  $C = 0.1$  and one with  $C = 1000$ . Which statements are necessarily true?

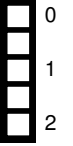
- ☒ The  $C = 0.1$  model has a wider margin (larger  $\frac{2}{\|\mathbf{w}\|}$ ).
- ☐ The  $C = 1000$  model has fewer support vectors.
- ☐ The  $C = 0.1$  model has smaller total slack  $\sum_i \xi_i^*$ .
- ☐ Increasing  $C$  beyond 1000 will eventually achieve 100% training accuracy on this dataset.
- ☐ The  $C = 1000$  model will always have higher test accuracy than the  $C = 0.1$  model.

### Problem 8 Singular Value Decomposition (5 credits)

A matrix  $\mathbf{M} \in \mathbb{R}^{4 \times 3}$  records the ratings of 4 users for 3 movies. Its SVD is given by  $\mathbf{M} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$  with

$$\mathbf{U} = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \\ 1 & -1 & -1 \end{pmatrix}, \quad \mathbf{\Sigma} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \quad \mathbf{V}^T = \frac{1}{3} \begin{pmatrix} 2 & 1 & -2 \\ 2 & -2 & 1 \\ 1 & 2 & 2 \end{pmatrix}.$$

a) Derive the best rank-2 matrix approximation of  $\mathbf{M}$  in terms of the Frobenius norm. You do not need to compute all entries and can instead derive its construction from  $\mathbf{U}$ ,  $\mathbf{\Sigma}$ , and  $\mathbf{V}$ .



The singular values on the diagonal of  $\mathbf{\Sigma}$  are  $\sigma_1 = 2$ ,  $\sigma_2 = 1$ ,  $\sigma_3 = 3$ . These are not sorted in decreasing order. The two largest are  $\sigma_3 = 3$  and  $\sigma_1 = 2$ , corresponding to the 3rd and 1st singular triplets. By the Eckart–Young theorem, the best rank-2 approximation retains the two largest singular values and their associated left/right singular vectors:

$$\mathbf{M}_2 = \sigma_3 \mathbf{u}_3 \mathbf{v}_3^T + \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T$$

where  $\mathbf{u}_i$  and  $\mathbf{v}_i$  are the  $i$ -th columns of  $\mathbf{U}$  and  $\mathbf{V}$ , respectively. Equivalently,  $\mathbf{M}_2 = \mathbf{U}_2 \mathbf{\Sigma}_2 \mathbf{V}_2^T$  with

$$\mathbf{U}_2 = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ -1 & 1 \\ 1 & 1 \\ -1 & 1 \end{pmatrix}, \quad \mathbf{\Sigma}_2 = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}, \quad \mathbf{V}_2^T = \frac{1}{3} \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \end{pmatrix}.$$

- 0 ☐ b) A new user rates the three movies as  $\mathbf{x} = (0, 3, 0)^T$ , where a rating of 0 indicates an unseen movie. Compute the representation of this user in the 2-dimensional concept space defined by the rank-2 approximation.

1 ☐

To project a new data point into the concept space, we use the right singular vectors  $\mathbf{V}_2$  (not  $\mathbf{U}$ , which encodes the original users):

$$\mathbf{z} = \mathbf{V}_2^T \mathbf{x} = \frac{1}{3} \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \end{pmatrix} \begin{pmatrix} 0 \\ 3 \\ 0 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 0+6+0 \\ 0+3+0 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

- 0 ☐ c) Reconstruct the predicted ratings for this user from the concept-space representation. Among the movies the user has not yet seen, which one does the model recommend?

1 ☐

2 ☐

$$\hat{\mathbf{x}} = \mathbf{V}_2 \mathbf{z} = \frac{1}{3} \begin{pmatrix} 1 & 2 \\ 2 & 1 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 2+2 \\ 4+1 \\ 4-2 \end{pmatrix} = \begin{pmatrix} \frac{4}{3} \\ \frac{5}{3} \\ \frac{2}{3} \end{pmatrix}.$$

The user has not seen movies 1 and 3 (original ratings 0). The predicted ratings are  $\frac{4}{3} \approx 1.33$  for movie 1 and  $\frac{2}{3} \approx 0.67$  for movie 3. The model recommends **movie 1**.

## Problem 9 GMMs & EM (6 credits)

a) Consider a GMM with  $K$  components where all components share the same covariance matrix  $\Sigma = \sigma^2 \mathbf{I} \in \mathcal{R}^{D \times D}$ . Let  $\gamma_{ik}$  denote the responsibility of component  $k$  for data point  $\mathbf{x}_i$ , computed in the E-step. Prove that in the limit  $\sigma \rightarrow 0$ , the responsibilities  $\gamma_{ik}$  converge to hard assignments, where each data point is assigned entirely to the component with the nearest mean.

*Hint:* Consider the ratio  $\frac{\gamma_{ik^*}}{\gamma_{ik}}$  where  $k^* = \arg \min_j \|\mathbf{x}_i - \mu_j\|^2$  and  $k \neq k^*$ .

The responsibility of component  $k$  for data point  $\mathbf{x}_i$  is given by

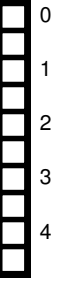
$$\gamma_{ik} = \frac{\pi_k \mathcal{N}(\mathbf{x}_i | \mu_k, \sigma^2 \mathbf{I})}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x}_i | \mu_j, \sigma^2 \mathbf{I})}.$$

Let  $k^* = \arg \min_j \|\mathbf{x}_i - \mu_j\|^2$ . For any  $k \neq k^*$ , the ratio of responsibilities is

$$\frac{\gamma_{ik^*}}{\gamma_{ik}} = \frac{\pi_{k^*}}{\pi_k} \exp \left( -\frac{\|\mathbf{x}_i - \mu_{k^*}\|^2 - \|\mathbf{x}_i - \mu_k\|^2}{2\sigma^2} \right).$$

Since  $\|\mathbf{x}_i - \mu_{k^*}\|^2 < \|\mathbf{x}_i - \mu_k\|^2$  by definition of  $k^*$ , the exponent is positive and diverges as  $\sigma \rightarrow 0$ , i.e.  $\frac{\gamma_{ik^*}}{\gamma_{ik}} \rightarrow \infty$ . Combined with  $\sum_k \gamma_{ik} = 1$  and  $\gamma_{ik} \geq 0$ , this implies  $\gamma_{ik} \rightarrow 0$  for all  $k \neq k^*$ . Thus, we conclude that in the limit

$$\gamma_{ik} \rightarrow \begin{cases} 1 & \text{if } k = \arg \min_j \|\mathbf{x}_i - \mu_j\|^2 \\ 0 & \text{otherwise.} \end{cases}$$



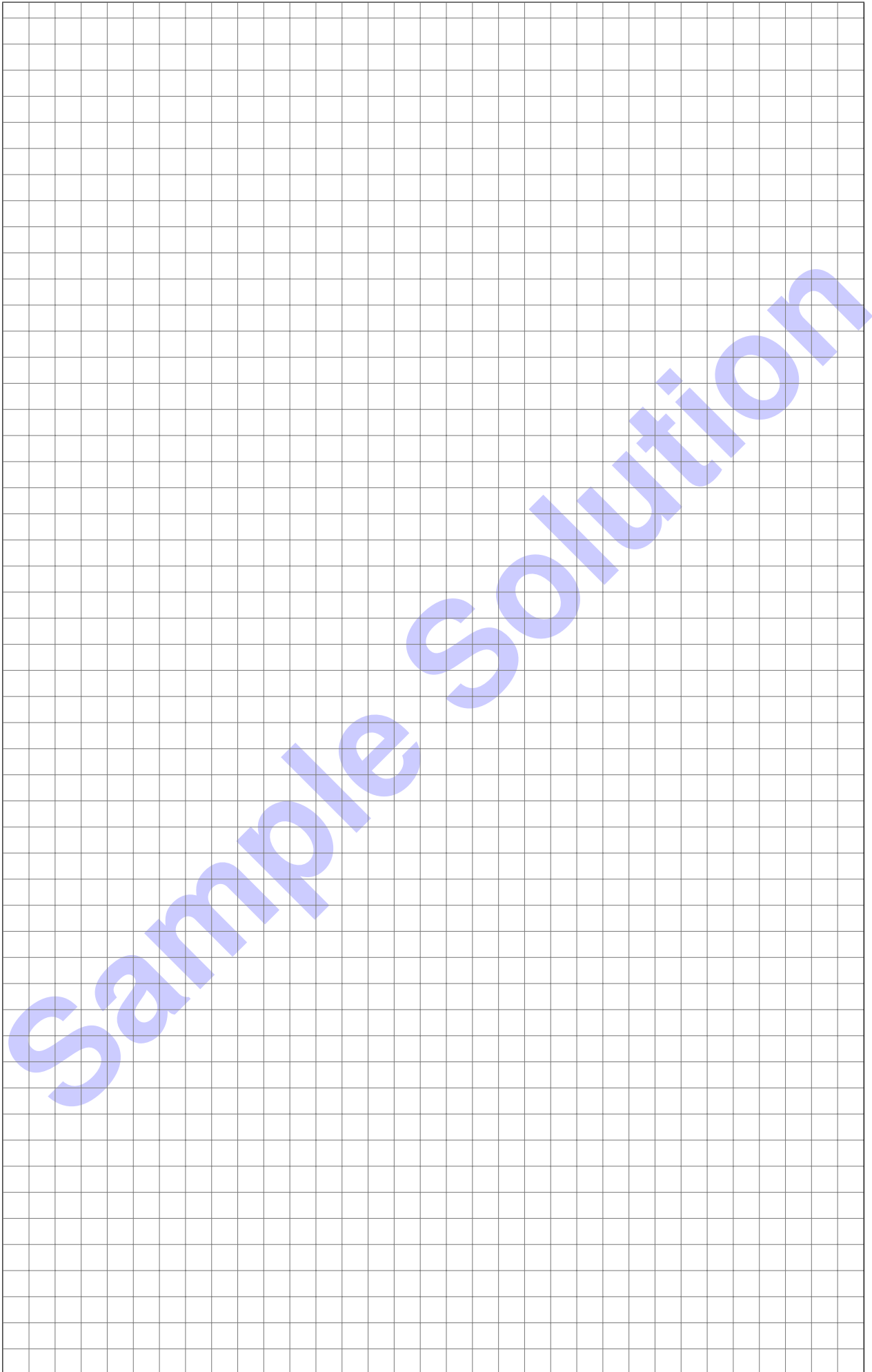
b) How does this change the M-step? Explain how this limit case relates to the K-means algorithm. Justify your answer.

This is exactly the assignment step in K-means, where each point is assigned to its nearest centroid. The subsequent M-step update for the means becomes  $\mu_k = \frac{1}{n_k} \sum_{i: \gamma_{ik}=1} \mathbf{x}_i$ , which is the K-means centroid update. Thus, K-means can be viewed as a limiting case of GMM-EM with isotropic covariances as  $\sigma \rightarrow 0$ .



Additional space for solutions—clearly mark the (sub)problem your answers are related to and strike out invalid solutions.

A large grid of graph paper, consisting of 30 columns and 30 rows of small squares. A diagonal watermark with the text "Sample Solution" in a light blue, sans-serif font is overlaid across the grid, running from the bottom-left towards the top-right.



Sample Solution